MATH 152 **EXAM I** Name Solution

Mrs. Bonny Tighe 6.1-7.5

50 points

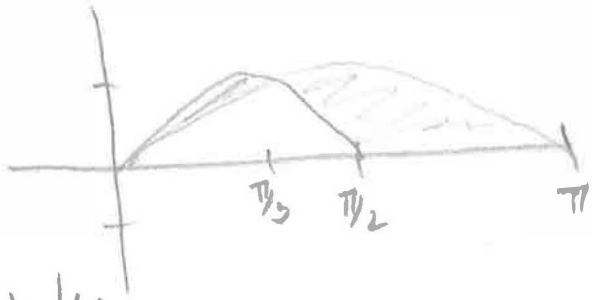
PART ONE TA_____Wed 10/4/2023

1. (11 points) Find the area of the region bounded the curves $y = \sin x$, *and* $y = \sin 2x$ on the interval $0 \le x \le \pi$. Graph.

$\sin x = 2\sin x \cos x$

$0 = \sin x\,(2\cos x - 1)$

$\sin x = 0 \quad \cos x = 1/2 \quad x = \pi/3$



$$\int_0^{\pi/3} (\sin 2x - \sin x)\,dx + \int_{\pi/3}^{\pi} (\sin x - \sin 2x)\,dx$$

$$-\tfrac{1}{2}\cos 2x + \cos x \Big|_0^{\pi/3} + -\cos x + \tfrac{1}{2}\cos 2x \Big|_{\pi/3}^{\pi}$$

$$\left(-\tfrac{1}{2}\cos\tfrac{2\pi}{3} + \cos\pi/3\right) - \left(-\tfrac{1}{2}+1\right) + \left(-\cos\pi + \tfrac{1}{2}\cos 2\pi\right) - \left(-\cos\pi/3 + \tfrac{1}{2}\cos\tfrac{2\pi}{3}\right)$$

$$-\tfrac{1}{2}(-1/2) + 1/2 - 1/2 \qquad -(-1) + \tfrac{1}{2}(1) + 1/2 - \tfrac{1}{2}(-1/2)$$

$$1 + 1/2 + 1/2 + 1/4$$

$$\tfrac{1}{4} \qquad 2\tfrac{1}{2}$$

2. (3 points each) True or false Circle the correct True or False

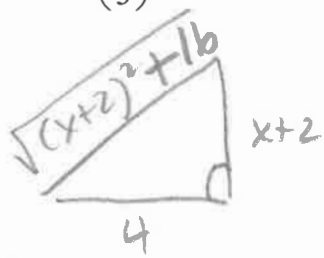
True False a) The functions $y = \tan^2 x$ and $y = \sec^2 x$ only vary by a constant.

True False b) $\int \frac{x^2+1}{x^2-4}\,dx = \int \frac{A}{x-2} + \frac{B}{x+2}\,dx$

True False c) $\int \frac{y^2}{\sqrt{4y^2-16}}\,dy$, we need to use the trig substitution $y = 2\sec\theta$.

True False d) The area between two curves could have a solution of $-\ln\left(\frac{2}{3}\right)$.

True False e) If $\tan\theta = \frac{(x+2)}{4}$, then $\sec\theta = \frac{\sqrt{(x+2)^2 - 16}}{4}$



True False f) To evaluate $\int \tan^2(w)dw$ you need to use u-substitution.

True False g) $\frac{1}{1-\cos x} = 1 - \sec x$.

True False h) $\int x^2 \sec(x^3)\,dx$ has to be evaluated using parts.

True False i) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^2(x+1)}$ would be $\frac{Ax+B}{x^2} + \frac{C}{x+1}$

True False j) The formula for the average value of a function is $f_{avg} = \int_a^b f(x)dx$

3. (3 points each) Multiple Choice. Circle the correct letter

a) How many terms does the partial fraction decomposition of $\frac{1}{x^6 + x^4}$ have.

$x^4(x^2+1)$

A) 7

B) 6

C) 1

D) 5

E) Some other answer

b) What method should not be used to integrate $\int \frac{x}{x^2-4}dx$

A) Trigonometric substitution

B) Partial Fraction Decomposition

C) Integration by Parts

D) U-Substitution

c) If the average value of the function $f(x) = 2x - 9x^2$ on the interval $[0, b]$ is -2, what is b?

A) 3

B) 1

C) $3/2$

D) 0

E) Some other answer

$\frac{1}{b}\int_0^b (2x-9x^2)dx = -2$

$\frac{1}{b}[x^2-3x^3]_0^b = \frac{1}{b}[b^2-3b^3]$

$b-3b^2-2=0$

$0=3b^2-b-2$

$0=(3b+2)(b-1)$

d) After making a substitution for $\int \frac{x^2\,dx}{\sqrt{16+x^2}}$ you get

A) $\int 16\sin^2\theta\cos\theta\,d\theta$

B) $\int 16\tan^2\theta\sec\theta\,d\theta$

C) $\int 16\tan^2\theta\,d\theta$

D) $\int \frac{1}{\sqrt{u}}du$, where $u = 16 + x^2$

E) Some other answer

$x = 4\tan\theta$

$dx = 4\sec^2\theta\,d\theta$

$\int \frac{16\tan^2\theta\ 4\sec^2\theta\,d\theta}{\sqrt{16+16\tan^2\theta}}$

$\int \frac{16\tan^2\theta\ 4\sec^2\theta\,d\theta}{\sqrt{16\sec^2\theta}}$

$\int 16\tan^2\theta\sec\theta\,d\theta$

e) Using the Cylindrical Shells method to find the volume of the solid of revolution for the function $y = 4x - x^2$ about the line $x = 4$, you would have:

A) $2\pi \int_0^4 (4-x)(4x - x^2 + 4)dx$

B) $2\pi \int_0^4 (x-4)(4x - x^2)dx$

C) $2\pi \int_0^4 (4-x)(4x - x^2)\,dx$

D) $2\pi \int_0^4 (x-4)(4x - x^2 - 4)\,dx$

E) Some other answer

f) On which of the following integrals can you NOT use the method of partial fractions ?

A) $\int_0^1 \frac{1}{x^2 - 16}dx$

B) $\int_0^1 \frac{1}{x^3 + 16}dx$

C) $\int_0^1 \frac{1}{x^2 + 16}dx$

D) $\int_0^1 \frac{1}{x^4 - 16}dx$

E) You can use partial fractions for all of the above.

MATH 152
Mrs. Bonny Tighe

EXAM I

6.1-7.5
50 points
PART TWO

NAME Solutions

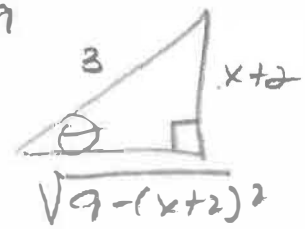
TA _____ Wed10/4/2023

1. (10 points) Integrate $\int \frac{(x+2)^2}{\sqrt{5-x^2-4x}}\,dx$

$-(x^2+4x+4)+5+4$
$-(x+2)^2+9$

$$\int \frac{(x+2)^2}{\sqrt{9-(x+2)^2}}\,dx$$

$(x+2) = 3\sin\theta$
$dx = 3\cos\theta\, d\theta$



$$\int \frac{9\sin^2\theta\, 3\cos\theta\, d\theta}{\sqrt{9-9\sin^2\theta}} = \int \frac{9\sin^2\theta\, 3\cos\theta\, d\theta}{\sqrt{9\cos^2\theta}} =$$

$$9\int \sin^2\theta\, d\theta = 9\int\left(\tfrac{1}{2}-\tfrac{1}{2}\cos 2\theta\right)d\theta$$

$$\tfrac{9}{2}\theta - \tfrac{9}{4}\sin 2\theta + C = \tfrac{9}{2}\theta - \tfrac{9}{2}\sin\theta\cos\theta + C$$

$$9/2\,\sin^{-1}\left(\tfrac{x+2}{3}\right) - \tfrac{9}{2}\left(\tfrac{x+2}{3}\right)\left(\tfrac{\sqrt{9-(x+2)^2}}{3}\right) + C$$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $x = 4y - y^2$ *and* $y = x$ rotated about the line y = 4. Graph

$y = 4y - y^2$
$0 = 3y - y^2$
$y = 0, 3$

$$2\pi\int_0^3 (4-y)\left[4y-y^2-y\right]dy =$$

$$2\pi\int_0^3 (4-y)(3y-y^2)\,dy = 2\pi\int_0^3 12y - 4y^2 - 3y^2 + y^3$$

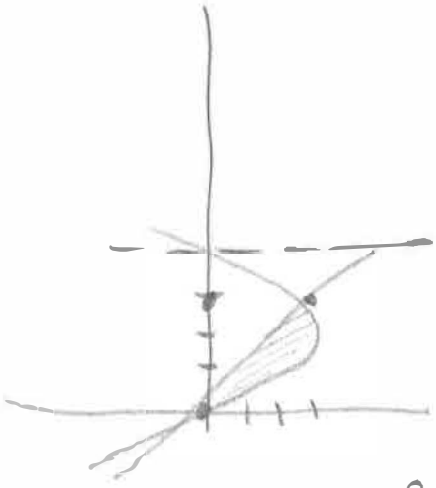
$$2\pi\int_0^3 (12y - 7y^2 + y^3)\,dy =$$

$$2\pi\left[6y^2 - \tfrac{7}{3}y^3 + \tfrac{1}{4}y^4\right]_0^3 =$$

$$2\pi\left[6(9) - \tfrac{7}{3}(27) + \tfrac{1}{4}(81)\right] - 0$$

$$2\pi\left[54 - 63 + \tfrac{81}{4}\right] = 2\pi\left[-9 + \tfrac{81}{4}\right]$$

$$2\pi\left[\tfrac{-36+81}{4}\right] = \frac{45\pi}{2}$$



3. (10 points) Integrate: $\int \cos(\ln t)\, dt$

$u = \cos(\ln t)$ $\quad dv = dt$

$du = -\sin(\ln t)\frac{1}{t}dt$ $\quad v = t$

$t\cos(\ln t) - \int t(-\sin(\ln t))\frac{1}{t}dt =$

$t\cos(\ln t) + \int \sin(\ln t)\, dt$

$u = \sin(\ln t)$ $\quad dv = dt$

$du = \cos(\ln t)\frac{1}{t}dt$ $\quad v = t$

$t\cos(\ln t) + \left[t\sin(\ln t) - \int t\cos(\ln t)\frac{1}{t}dt\right]$

$t\cos(\ln t) + t\sin(\ln t) - \int \cos(\ln t)\, dt$ ☺

2☺

$\int \cos(\ln t)dt = \frac{1}{2}\left[t\cos(\ln t) + t\sin(\ln t)\right] + C$

need more detail

4. (10 points) Integrate: $\int_{\sqrt{3}}^{3} \frac{x^3}{\sqrt{9+x^2}}\, dx$

$x = 3\tan\theta$

$dx = 3\sec^2\theta\, d\theta$

$\tan\theta = x/3$

$\tan\theta = \sqrt{3}/3$

$\theta = \pi/6$

$\tan\theta = 3/3 = 1$

$\theta = \pi/4$

$\int_{\pi/6}^{\pi/4} \frac{27\tan^3\theta\, 3\sec^2\theta\, d\theta}{\sqrt{9+9\tan^2\theta}}$

$\int_{\pi/6}^{\pi/4} \frac{27\tan^3\theta\, 3\sec^2\theta\, d\theta}{3\sec\theta} = 27\int_{\pi/6}^{\pi/4} \tan^3\theta\sec\theta\, d\theta$

$\tan^2\theta\sec\theta\tan\theta\, d\theta$

$(\sec^2\theta - 1)\sec\theta\tan\theta\, d\theta$

$u = \sec\theta$

$du = \sec\theta\tan\theta\, d\theta$

$27\int_{\pi/6}^{\pi/4} (\sec^2\theta - 1)\sec\theta\tan\theta\, d\theta$

$27\left[\frac{1}{3}\sec^3\theta - \sec\theta\right]_{\pi/6}^{\pi/4}$

$27\left[\left(\frac{1}{3}(\sec\pi/4)^3 - \sec\pi/4\right) - \left(\frac{1}{3}(\sec\pi/6)^3 - \sec\pi/6\right)\right]$

$27\left[\frac{1}{3}(\sqrt{2})^3 - \sqrt{2} - \frac{1}{3}\left(\frac{2}{\sqrt{3}}\right)^3 + \frac{2}{\sqrt{3}}\right]$

5. (10 points) Integrate $\int \frac{x^5+2}{x^4+x^2}\,dx$

$$\begin{array}{r} x \\ x^4+x^2 \overline{\big) x^5 \quad +2} \\ -(x^5+x^3) \\ \hline -x^3+2 \end{array}$$

$$\int x + \frac{-x^3+2}{x^2(x^2+1)}\,dx$$

$$\frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} = \frac{-x^3+2}{x^2(x^2+1)}$$

$$Ax(x^2+1) + B(x^2+1) + x^2(Cx+D) = -x^3+2$$

$x=0$ $\quad B=2$

$$Ax^3+Ax+Bx^2+B+Cx^3+Dx^2 = -x^3+2$$

$$x^3(A+C) + x^2(B+D) + Ax + B$$

$A+C=-1$ $\quad D=-2$ $\quad A=0$

$$\int \left(x + 2x^{-2} - \frac{x}{x^2+1} - \frac{2}{x^2+1}\right)dx$$

$$\frac{1}{2}x^2 - \frac{2}{x} - \frac{1}{2}\ln|x^2+1| - 2\tan^{-1}x + C$$

6. (5 points) Integrate $\int \tan^{-1}\sqrt{x}\,dx$

$m=\sqrt{x}$

$dm = \frac{1}{2\sqrt{x}}dx \qquad dx = 2m\,dm$

$$\int 2m\tan^{-1}m\,dm$$

$u=\tan^{-1}m \qquad dv = 2m\,dm$

$du = \frac{1}{1+m^2}dm \qquad v=m^2$

$$m^2\tan^{-1}m - \int \frac{m^2}{1+m^2}dm = m^2\tan^{-1}m - \int 1 - \frac{1}{1+m^2}dm$$

$$m^2\tan^{-1}m - m + \tan^{-1}m + C$$

$$\boxed{x\tan^{-1}\sqrt{x} - \sqrt{x} + \tan^{-1}\sqrt{x} + C}$$